

KAVEESHA DHANANJAYA

+94740 179 243 | kaveeshadhananjaya2002@gmail.com | <https://kd-27.github.io/portfolio/>

I am a dedicated Mechatronics Engineering student with a strong academic record and hands-on experience in robotics and automation. Skilled in mechanical design, electrical modeling, and programming, with expertise in designing and fabricating robotic systems for real-world applications. Proficient in advanced modeling tools like SolidWorks and MATLAB, with hands-on experience in programming and integrating PLCs for industrial automation. Recognized for leadership roles in academic and extracurricular settings, I bring a collaborative mindset and a passion for innovative technology solutions. Eager to contribute to dynamic engineering projects that drive automation and intelligent design.

EDUCATION

BSc Eng (Hons) in Mechatronics Engineering, **2021 – 2025**
General Sir John Kotelawala Defense University, Ratmalana

- Cumulative GPA: 3.91 / 4.0
- Key modules: Advanced Robotics, Machine Elements in Design, Power Electronics, Digital Signal Processing

Primary & Secondary Education, Royal College, Colombo **2007 – 2020**

- Z- Score : 1.4 | Physical Science

WORK EXPERIENCE

Senior Robotics Engineer, Hype Insight (Pty) Ltd **9/2025 – present**

- Design and development of four wheel robot integrating AI for computer vision.
- Research and development of Rover and drone concepts .
- Building Mic array by developing an audio processing pipe-line.

Product Development Engineer - Innovations, MAS Capital (Pvt) Ltd, Biyagama **4/2025 – 8/2025**

- Fabrication and Development of Mechanical systems to enhance process Efficiency.
- Research and Innovation on new technologies to Develop Products .
- Feasibility Analysis to develop new Concepts.

Tutor - 360 Labs, Borelasgamuwa (Hybrid) **12/2023 – 12/2024**

- Designed STEM kits and led workshops to engage students aged 5–20 in STEM concepts.
- Delivered robotics courses in schools to spark interest in technology.
- Completed automation projects to advance agricultural practices.

Intern (Autonomation Engineer) - Autonomation Lab, MAS Capital (Pvt) Ltd, Ratmalana **10/2023 – 03/2024**

- Assisted in the design, fabrication, testing, validation, and documentation processes of new developments.
- Designed and fabricated three modular units using PLC for strap making, ironing and cutting.
- Developed a Convolution Neural Network to Identify the presence and orientation of a bra cup.

PROJECT EXPERIENCE

Quadruped Robot for Inspection in Human-Inaccessible Structured Areas, KDU, Ratmalana **Ongoing**

- Four-legged robot equipped with a five-bar mechanism for each leg to navigate structured terrains.
- Responsibilities: MATLAB simulation, ROS2 communication, wiring diagrams, inverse kinematic modeling

Mars Rover-Inspired Wheeled Robot with LiDAR for Mapping and Navigation, KDU, Ratmalana **2023**

- Designed and fabricated a Mars Rover-like wheeled robot equipped with LiDAR which utilizes SLAM algorithm on ROS to map, localize, and navigate in a structured environment.
- Responsibilities: Kinematics modeling, GAZEBO simulation, 3D Modeling, Electronics Assembly

Sweep the Table Robot Contest, KDU, Ratmalana **2023**

- Designed a wheeled robot to autonomously push objects off a rectangular arena using a rotating hammer.
- 1st place and "Best Design" award at KDU FOE Intake 38 Mechatronics Systems Design Competition.
- Responsibilities: Developing pushing mechanisms Electronics Assembly, Fabrication, Programming

PUBLICATIONS

Design & Development of a Quadruped Robot for Inspection in Human Inaccessible Structured Areas • Publication Under Progress	2025
Prediction of Coronary Artery Disease using Artificial Neural Network • KDU- 17th International Research Conference	2024
Development of an Automated Clothesline System • KDU- 16th International Research Conference	2023

EXTRACURRICULAR ACTIVITIES

Research In-charge , Electronic Robotics and Innovation Club, KDU, Ratmalana	12/2022 – 12/2023
Senior Prefect , Royal College, Colombo	12/2020 – 12/2021
President Scout , Royal College, Colombo	12/2022
Captain, Football , Royal College, Colombo	2014, 2016, 2017

CERTIFICATIONS AND TRAININGS

Machine Learning , DeepLearning.AI, Stanford University - Coursera	07/2023 – 08/2023
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SKILLS AND LANGUAGES

PLC & HMI Programming: ISPSoft, SIMATIC STEP7, SKTool, SATool, Nextion IDE
Mechanical Modeling and Designing: SolidWorks, Fusion360, Ansys, MATLAB
Microcontrollers and processors: SolidWorks, Fusion360, Ansys, MATLAB
Simulation Software: MATLAB (Simulink, Simscape) , GAZEBO(ROS2 Environment)
Electrical Modeling and Designing: Proteus, EasyEDA, KiCAD, LTSpice, Eagle
Programming Languages: Python, C++
Languages: English (Fluent), Sinhala (Native)

HOBBIES AND INTERESTS

RC Car Racing, Football, 3D Printing

REFERENCE

Mr. Isira Naotunna

BSc. Hons(Moratuwa, Sri Lanka), M.Eng(CMU, Thailand)

Senior Lecturer, Department of Mechanical Engineering,

General Sir John Kotelawala Univeristy, Kandawala Road, Dehiwala-Mount Lavinia, Sri Lanka

Phone: +94 71 627 8136

Email: isira.naotunna@adelaide.edu.au

Mr. H M D L Dabare

Senior Automation Engineer

Automation Department,

MAS Capital (Pvt) Ltd - Intimates Division, 7th Lane, Off Borupana Road, Ratmalana, Sri Lanka

Phone: +94 77 406 1725

Email: mirand@masholdings.com